Caoliwen Wang

wclw1021@gmail.com | wclw1021.github.io

Research Interests

Rooted in computer vision and graphics, scientific computing, and generative AI, my research develops **physically grounded, scalable, and generative models** for physical world simulation and video/3D content generation.

Education

University of British Columbia Sep 2025 - Present
Ph.D. in Computer Science
Advisor: Prof. Peter Yichen Chen

University of Science and Technology of China Sep 2021 - Jun 2025
B.S. in Computational Mathematics
Advisor: Prof. Ligang Liu

Research Experience

Scalable Physical Simulation Jun 2025 - Present

- **WorldParticle**: Developed a unified model that represents diverse physical phenomena as Lagrangian particles and learns their dynamics with a Transformer-based backbone, enabling fully data-driven simulation without the hand-crafted design and pre-defined physical biases required by traditional solvers, toward unified and scalable world modeling.
- **WorldParticle++**: Built upon the WorldParticle backbone, scaling particle-based simulation to million-scale systems with support for dynamic boundaries, while leveraging forcing techniques from video-based diffusion models to improve rollout stability for long-horizon simulation.
- **Agentic Simulation**: Developed a fully automated simulation pipeline leveraging LLM agents for 3D scene construction and solver configuration, eliminating tedious manual parameter tuning, making large-scale 3D data generation feasible, and enabling integration with WorldParticle toward a unified 3D Physics Foundation Model.
- **Agentic Sim-to-Real**: Leveraged LLM agents to automatically close the sim-to-real gap for robotic manipulation of diverse types of powders, generating scalable data for training policies that generalize across different powder materials.

Physics Informed Content Generation Jun 2025 - Present

- **Material-aware Video Generation**: Fine-tuned Wan2.2 with multi-material and non-interaction video data, demonstrating that material-aware training can significantly enhance video foundation models' ability to generate videos involving interactions among diverse materials.
- **Physically Plausible 3D Configuration Generation**: Developed a method that enforces physical validity, including non-penetration and structural plausibility, on generated 3D protein configurations, improving the inference efficiency of protein structure prediction foundation models.

Design-Driven 3D Shape Generation Jan 2024 - Jun 2025

- **Neural Shadow Art**: Developed an implicit neural 3D representation that facilitates geometry generation from multi-view binary shadow targets, with angular adjustment and volume/geometry optimization.
- **Shape from Semantics**: Developed a 3D generation framework for creating semantically consistent shapes whose multi-view renderings align with user-specified prompts, extending the inverse-modeling framework to a semantics-conditioned 3D generation problem.

Industry Experience

Style3D Jun 2026 - Present

Scaled WorldParticle to human motion and cloth grasping scenarios, demonstrating its ability to generalize across diverse human motions and cloth manipulation strategies.
Mentor: Prof. Huamin Wang

Publications & Preprints

[1] **Caoliwen Wang***, Minghao Guo*†, Siyuan Chen*, Heng Zhang, Mengdi Wang, Xingyu Ni, Hanson Sun, Kunyi Wang, Zherong Pan, Kui Wu, Lingjie Liu, Yin Yang, Chenfanfu Jiang, Taku Komura, Wojciech Matusik, and Peter Yichen Chen†. WorldParticle: Unified World Simulation of Lagrangian Particle Dynamics via Transformer. *SIGGRAPH Asia (Conference Track), 2026.*

[2] Xingyu Ni†, Jiong Chen, Siyuan Chen, **Caoliwen Wang**, Mathieu Desbrun, and Taku Komura. Efficient Open-Boundary Poisson Solves. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia), 2026.*

[3] Siyuan Chen*, Minghao Guo*†, **Caoliwen Wang**, Anka He Chen, Yikun Zhang, Jingjing Chai, Yin Yang, Wojciech Matusik, and Peter Yichen Chen†. Physically Valid Biomolecular Interaction Modeling with Gauss-Seidel Projection. *International Conference on Learning Representations (ICLR), 2026.*

[4] Liangchen Li, **Caoliwen Wang**, Yuqi Zhou, Bailin Deng, and Juyong Zhang†. Shape from Semantics: 3D Shape Generation from Multi-View Semantics. *Preprint, Under Review at TVCG, 2025.*

[5] **Caoliwen Wang**, Bailin Deng, and Juyong Zhang†. Neural Shadow Art. *Pacific Graphics (Conference Track), 2025.*

Awards

University and Provincial Outstanding Graduate Award 2025 (USTC)

Reviewing

ACM SIGGRAPH Asia, Pacific Graphics